

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1478

Los Molinos 138 kV
April 26, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1478. The Interconnection Customer proposes a 170.9 MW solar photovoltaic facility interconnecting at the Los Molinos 138 kV substation in the Virginia area. The request was received on May 16, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 385 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,056,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Carver to Big Hanaford 138 KV L
Contingency: Plaza Substation to Big Hanaford Substation 138 KV Line
Loading: 118.1% (Rating: 264 MVA)
DFAX: 25.2%
Pre-project loading: 101.8%
2. Facility: Shockoe to Carver 138 KV
Contingency: Plaza to Big Hanaford 138 KV
Loading: 112.4% (Rating: 344 MVA)
DFAX: 25.2%
Pre-project loading: 99.8%
3. Facility: Fritch to Big Hanaford Substation 138 KV
Contingency: Mitchell Dam to Skaggs Island Jct 2 345 KV line
Loading: 105.3% (Rating: 529 MVA)
DFAX: 32.5%
Pre-project loading: 94.7%
4. Facility: Manchester to Shockoe 138 KV
Contingency: Plaza to Big Hanaford 138 KV
Loading: 105.2% (Rating: 306 MVA)
DFAX: 9.6%
Pre-project loading: 99.8%
5. Facility: Plaza to Big Hanaford 138 KV line
Contingency: Mitchell Dam to Skaggs Island Jct 2 345 KV
Loading: 102.1% (Rating: 459 MVA)
DFAX: 16.1%
Pre-project loading: 96.2%

6. Facility: Turner Road to Tyler 138 KV
 Contingency: Mitchell Dam to Haggin 345 KV
 Loading: 101.1% (Rating: 392 MVA)
 DFAX: 10.4%
 Pre-project loading: 96.6%
7. Facility: Kimberley to Shack Mills Sta 138 KV
 Contingency: Chancellor Row to Bear Island 345 KV
 Loading: 100.5% (Rating: 249 MVA)
 DFAX: 7.8%
 Pre-project loading: 95.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carver to Big Hanaford 138 KV	Rebuild line with higher-capacity conductor	\$2,079,000	12% (\$251,000)
Shockoe to Carver 138 KV	Rebuild line with higher-capacity conductor	\$1,131,000	12% (\$136,000)
Fritch to Big Hanaford Substation 138 KV	Reconductor line	\$1,559,000	7% (\$105,000)
Shockoe Substation to Manchester 138 KV	Reconductor line	\$1,910,000	10% (\$194,000)
Plaza to Big Hanaford 138 KV line	Reconductor line	\$2,559,000	6% (\$147,000)
Turner Road to Tyler 138 KV	Reconductor line	\$2,864,000	6% (\$184,000)
Kimberley to Shack Mills Sta 138 KV	Reconductor line	\$632,000	6% (\$39,000)
	Total allocated to project		\$1,056,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.